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A New Blind Deblurring Method via Hyper-Laplacian Prior

Jun Kong[a,b,]* , Kesai Lu[a], Min Jiang[a]

[a]*Key Laboratory of Advanced Process Control for Light Industry Jiangnan University, No. 1800 Lihu Avenue, Wuxi 214122, China*
[b]*College of Electrical Engineering Xinjiang University, No. 14 Shengli Road, Urumqi 830046, China*
** Corresponding author：kongjun@jiangnan.edu.cn Tel.: (+86)180-1235-5422*

Abstract

Blind image deblurring is always a thorny problem due to its uncertainty. In this manuscript, we proposed a new blind deblurring method in the maximum a posterior (MAP) framework to remove the blur of the image. Firstly, we choose the hyper-Laplacian prior to be a regularization of the gradients of an image. Secondly, we adopt an operator called generalized soft thresholding (GST) to solve the non-convex problem during the whole deblurring process. Thirdly, we compared our method with some popular approaches in both quantitative and qualitative aspects. Experimental results show that the method proposed by us performs much better than the other approaches.

Keywords: Blind deblurring; hyper-Laplacian prior; generalized soft thresholding.

1. Instruction

During the process of using a digital camera to obtain images, the relative motion between the lens of camera and the scene often makes the image blurry. With the development of science and technology, the anti-shake or image stabilizer technique has been gradually used in cameras with high configuration and some intelligent mobile phones. So the image deblurring has great practical significance. In this paper, we mainly research how to remove the motion blur through the algorithm in mathematics and graphics but not using the special hardware. And we suppose the motion is caused by the camera shake while the scene and the object are static. This kind process of image degradation often called the blur in overall situation, and its degraded model can be calculated as

$$Y = X * k + n, \tag{1}$$

where Y represents the blurry image, X denotes the latent or deblurred image, k is called the blur kernel, it represents the motion trajectory of the equipment when exposing, n and $*$ represent the noise and the convolution operator, respectively.

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We usually divide the image deblurring into two classes, *i.e.*, non-blind and blind deblurring according to the kernel k is given and not known, respectively. In non-blind deblurring, k is already known or estimated by some way. So the remaining task is to restore the final latent image. In a real-word image, its gradients present the heavy-tailed distribution. Krishnan and Fergus[1] proposed a fast non-blind image deblurring method using hyper-Laplacian prior, *i.e.*, $f(a) \propto e^{-m|a|^p}$ $(0.5 \le p \le 0.8)$ to model this distribution. They have verified that such prior is superior to Laplacian and Gaussian priors. However, because of the $p < 1$ in the hyper-Laplacian prior the calculated model will become a puzzled non-convex problem. In their work, they mainly analyzed two specific fraction values, *i.e.*, $p = 1/2$ and $p = 2/3$ by calculating a cubic and quartic polynomial, respectively. And then they need to choose the appropriate roots and get rid of invalid results. But this kind method needs to compute different polynomials according to the different p. Besides, the complexity of polynomial will become harder with the increase of the denominator. So the researchers usually adopt the method, *i.e.*, iteratively reweighted least squares (IRLS) to deal with this problem. But IRLS often cannot converge to the global optimal solution. Zuo *et al.*[2] then proposed a GST operator for non-convex l_p $(0 \le p < 1)$ norm optimization problem. And the GST operator can easily deal with the non-convex sparse problems with the arbitrary p values. In most cases, blind deblurring is more common because of the unknown blur kernel. The target of blind deblurring is to obtain X and k with the given Y. The key task of the blind deblurring is to estimate the blur kernel, and the remaining work is to restore the final latent image through the non-blind deblurring method.

In this manuscript, a new blind deblurring method via hyper-Laplacian prior has been proposed. This prior is mainly used for restoring the final latent image in non-blind deblurring, while in our work we adopt it to update the intermediate latent image. Besides, the GST operator is utilized to deal with the non-convex problem caused by such prior. At last, when the blur kernel has been estimated, we use the method of updating the intermediate latent image to recover the final latent image. Experiments restored by our method prove much better than some popular methods.

2. Deblurring Algorithm

Blind image deblurring problem can be solved by alternately estimating the intermediate latent image and the blur kernel in an iterative process. In this section, we will introduce a new blind deblurring algorithm under this optimization scheme.

In the intermediate latent image estimation process, we use the hyper-Laplacian prior as a regularization and a GST operator to solve the proposed model. Then the estimated intermediate latent image is used to estimate the blur kernel in the next iteration. When the final blur kernel is obtained, a final latent image recovery process is implemented.

During estimating the kernel, the Gaussian prior is usually used to model the property of its distribution.

The details of this procedure are summarized in Fig. 1.

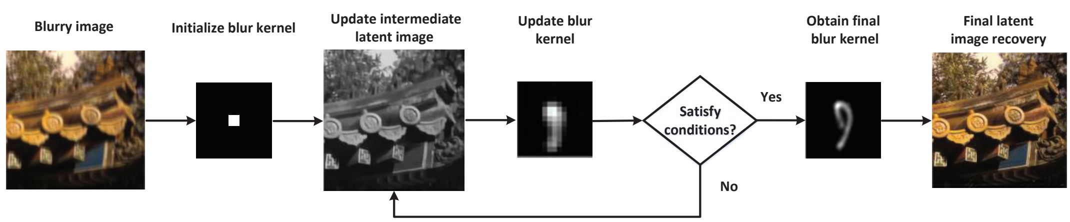

Fig. 1. The flowchart of our algorithm.

2.1. Estimating Intermediate Latent Image X

When we get k, then X can be calculated by

$$X^* = \arg\min_X \|X * k - Y\|_2^2 + \lambda \|\nabla X\|_p^p, \tag{2}$$

where $\nabla X = (\partial_h X, \partial_v X)^T$ is the horizontal and vertical gradient of the image X, λ is the weight of the regularization, $0.5 \le p \le 0.8$. Due to the hyper-Laplacian regularization term in (2), minimizing (2) is computationally intractable. We employ an efficient alternating minimization method[3] to solve this problem. We introduce the auxiliary variable $g = (g_h, g_v)^T$ with respect to the horizontal and vertical gradients of the image. Then we rewrite (2) as the following (3),

$$(X^*, g^*) = \arg\min_{X,g} \|X * k - Y\|_2^2 + \beta \|\nabla X - g\|_2^2 + \lambda \|g\|_p^p, \tag{3}$$

where β is the penalty parameter. And the result of (3) will approach that of (2) when β is close to infinity. Then we deal with (3) by alternatively updating X and g independently via making another variable fixed.

We firstly initialize g to be zero. In each iterative process, X will be acquired by (4),

$$X^* = \arg\min_X \|X * k - Y\|_2^2 + \beta \|\nabla X - g\|_2^2. \tag{4}$$

The solution of (4) can be calculated through the following (5),

$$X^* = F^{-1}\left(\frac{\overline{F(k)}F(Y) + \beta F_G}{\overline{F(k)}F(k) + \beta \overline{F(\nabla)}F(\nabla)} \right), \tag{5}$$

where $F(\bullet)$ and $F^{-1}(\bullet)$ represent the Fourier transform and its inverse transform, respectively; $\overline{F(\bullet)}$ denote the complex conjugate operation; $F_G = \overline{F(\partial_h X)}F(g_h) + \overline{F(\partial_v X)}F(g_v)$, $\overline{F(\nabla)}F(\nabla) = \overline{F(\partial_h X)}F(\partial_h X) + \overline{F(\partial_v X)}F(\partial_v X)$.

Given X, we compute g by

$$g^* = \arg\min_g \beta \|\nabla X - g\|_2^2 + \lambda \|g\|_p^p. \tag{6}$$

Lemma 2.1 *The generalized soft thresholding (GST) operator*

Let z be a single variable,

$$z^* = \arg\min_z \frac{1}{2}(z - y)^2 + \mu |z|^p, \tag{7}$$

the optimal solution z^* is defined as

$$z^* = \begin{cases} \operatorname{sgn}(y)\left(|y| - \mu p z_l^{p-1}\right), & |y| > \tau_p^{GST}(\mu) \\ 0, & otherwise \end{cases}, \tag{8}$$

where $\tau_p^{GST}(\mu)$ is the threshold decided by p and μ, z_l denotes the intermediate solution after the l th iteration.

According to *Lemma 2.1*, the solution of (6) can be solved by

$$g^* = \begin{cases} \operatorname{sgn}(\nabla X)\left(|\nabla X| - \lambda p g_l^{p-1}\right), & |\nabla X| > \tau_p^{GST}(\lambda) \\ 0, & otherwise \end{cases}. \tag{9}$$

Algorithm 1 shows the main steps to calculate (2).

Algorithm 1 Calculating (2)

Inputs: Y and k
1: $X \leftarrow Y$, $\beta \leftarrow 2\lambda$
2: **repeat**
3: obtain g by (9)
4: obtain X by (5)
5: $\beta \leftarrow 2\beta$
6: **until** $\beta > \beta_{\max}$
Output: X

2.2. Estimating Blur Kernel k

With the given X, the model of updating k is

$$k^* = \arg\min_k \|X * k - Y\|_2^2 + \gamma \|k\|_2^2. \tag{10}$$

We adopt the method of selecting the salient edges[4] from the intermediate latent image X to estimate the kernel k. Because the solution directly from (10) may acquire the delta kernel solution, which is an invalid solution[5]. So in order to avoid this situation, we rewrite (10) as

$$k^* = \arg\min_k \|\nabla S * k - \nabla Y\|_2^2 + \gamma \|k\|_2^2, \tag{11}$$

where ∇S denotes the selected salient edges of X. Because (11) is a least squares minimization problem, so we can solve it by FFTs,

$$k^* = F^{-1}\left(\frac{\overline{F(\partial_h S)} F(\partial_h Y) + \overline{F(\partial_v S)} F(\partial_v Y)}{F(\partial_h S)^2 + F(\partial_v S)^2 + \gamma} \right), \tag{12}$$

where $k(i,j) \geq 0$, and $\sum_i \sum_j k(i,j) = 1$.

We construct an image pyramid through a coarse to fine way to estimate the kernel k. When we begin to execute a pyramid level, we firstly need to initialize k with the intermediate latent image X from the previous coarser level. Here, we set the number of iterations between k and X in each pyramid level to be 5. After each iteration, we chose an adaptation strategy to update the variable λ. Algorithm 2 presents the main steps of estimating the blur kernel k on one pyramid level.

Algorithm 2 Estimating the kernel k

Input: Y

1: Initialize k from the previous coarser level
2: **for** $i = 1$ to 5 **do**
3: obtain X by Algorithm 1
4: obtain k by (11)
5: $\lambda \leftarrow \max\left\{\lambda / 1.1, 1e^{-4}\right\}$
6: **end for**

Outputs: X and k

2.3. Restore the Final Latent Image

After the final blur kernel k has been obtained, the next work is to restore the final latent image via a non-blind deblurring way. In our work, we recover the final latent image through (2). And here we need to restore the final image in grey-scale or RGB according to our requriments.

3. Experiments

In the experiments, we compare the results obtained by our approach with the blind deblurring methods of the references[4-8]. Firstly, some implementation details about experiments are introduced. We empirically set the following parameters $\lambda = 4e^{-3}$, $\gamma = 2$ and $\beta_{\max} = 1e^{5}$, respectively. The value of p is typically set within the range of $\left[0.5, 0.8\right]$. In our work, we conduct the abundant experiments and finally we set $p = 0.8$.

We test our method on the dataset[9]. The error metric is the ratio of sum of squared differences (SSD). In Fig.2, the cumulative histogram of error ratios across the dataset is shown. The values less than three are considered acceptable from the visual aspect. And the success rate will be higher when the value is lower, which indicates the result restored by the method is better. From the Fig.2, we can see that the result obtained by our method is much better than the other methods.

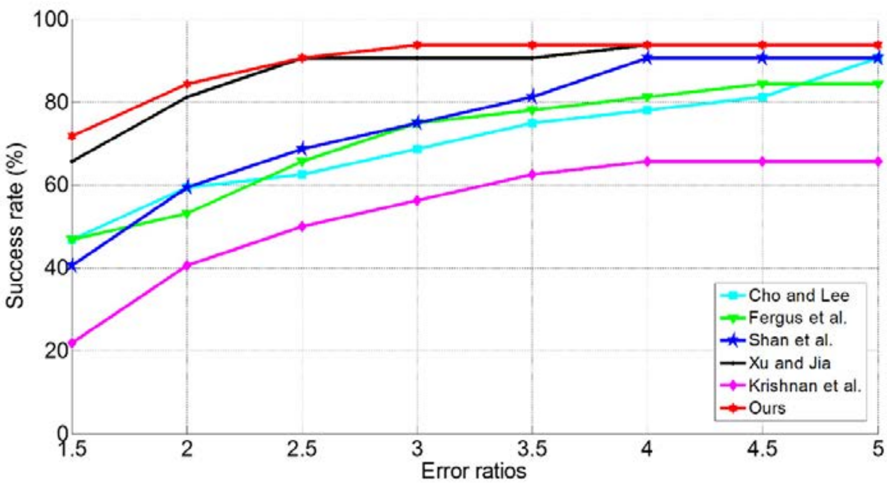

Fig. 2. Cumulative histogram of the error ratios across the dataset[9].

In addition, we test our method on another natural image deblurring dataset[10]. And we adopt a quantitative image assessment metric, *i.e.*, peak signal to noise ratio (PSNR) to assess the results. The higher the values, the better the results. From Fig. 3 we can see average PSNR values of each image are all the highest.

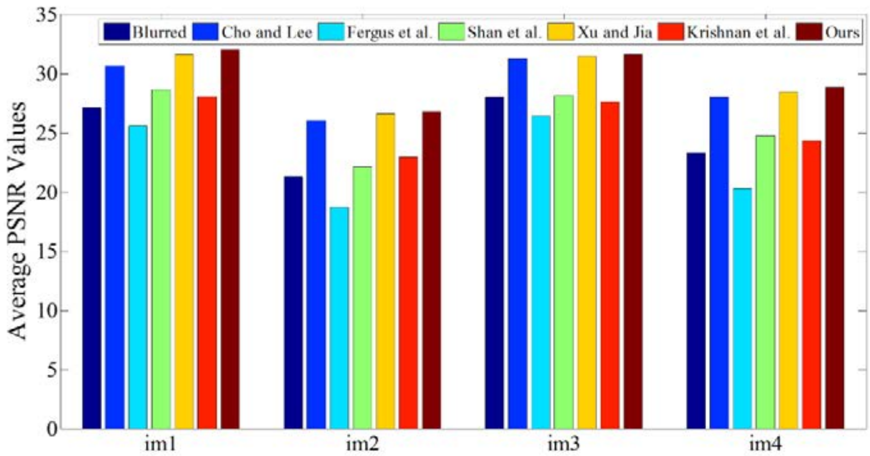

Fig. 3. Average PSNR values on each image of dataset[10].

We further test our method on an example. Fig. 4 presents the kernels and the final results deblurred by different methods. We can see the results of references[4,5,6] still contain some noise and visual artifacts. Besides, a representative region extracted with a red box is accordingly shown in Fig. 5. Ours result contain more details than references[7,8], so the result of ours is easier to be accepted by referring to the groundtruth.

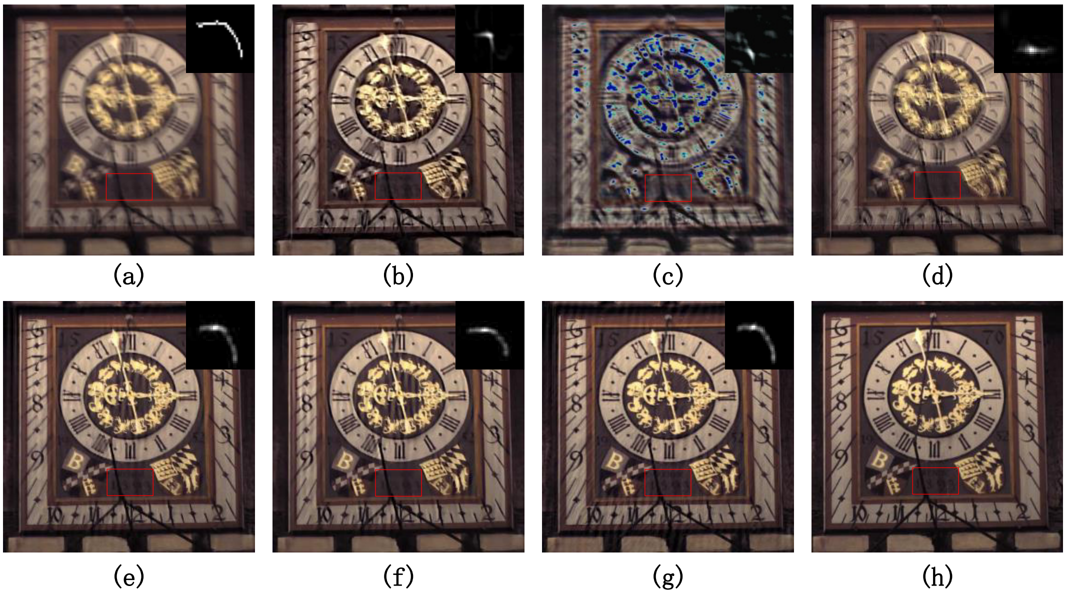

Fig. 4. An example restored by different methods.

(a) Blurred image and truth kernel; (b)-(g) represent the results of conferences[4-8] and ours. The PSNR values in (b)-(g) are 20.249, 25.295, 15.523, 19.572, 25.737, 20.037, and 26.012, respectively. Our method performs the best.

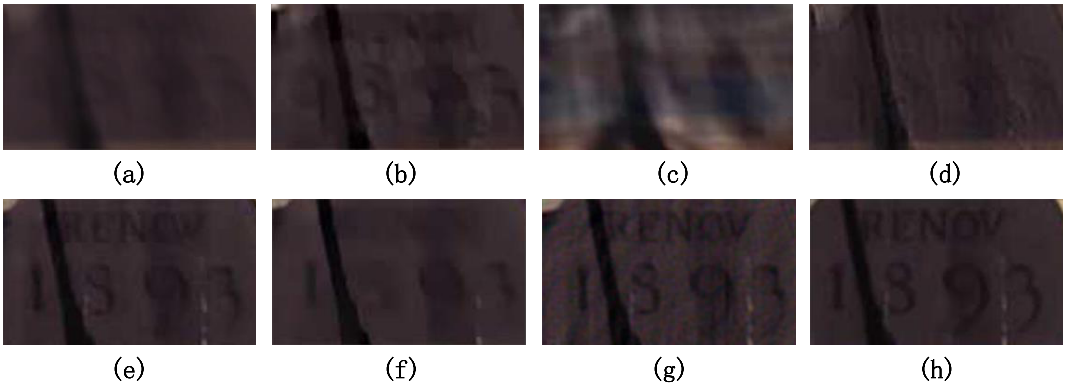

Fig. 5. The representative region are extracted with a red box according to Fig. 4, respectively.

4. Conclusions and Future Work

To sum up, we propose a new blind deblurring method via hyper-Laplacian prior. We mainly adopt this prior to estimate the intermediate latent image. And the GST operator is introduced to deal with the non-convex optimal problem caused by such prior. The usage of operator is not only flexible but also precise comparing to the other methods. The experiments prove that our method can estimate both the kernel and the image effectively. However the approach in this paper can only deal with the condition of spatially invariant blur kernel, so we will be committed to solving the spatially variant problem in the next research.

Acknowledgments

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